UNIVERSITAT POLITÈCNICA DE CATALUNYA
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Lab Session 9: Jitter and Latency Handling

Latency Handling

- Problems:
  - Laggy input response
  - Abrupt movement of entities
  - Non-expected reaction to inputs

Client-side Prediction

- This technique attempts to make the player's input feel more instantaneous while governing the player's actions on a remote server

- If the game is deterministic enough, instead of sending the inputs and waiting for the new game state to start rendering it, we can send the input and start rendering the outcome of that inputs as if they had succeeded, while we wait for the server to send the “true” game state.

Problem with client-side prediction

- Mismatch between the state perceived by the client (prediction) and the state forced by the server (“real”?)
- Discarding predictions in favor of the current state from server (replications) can lead to different perceptions of the actual events
- Solution?

Server reconciliation

- The client includes a sequence number in every input sent to the server, and keeps a local copy.
- When the server sends an authoritative update to a client, it includes the sequence number of the last processed input for that client.
- The client accepts the new state, and reapplies the inputs not yet processed by the server, completely eliminating visible desynchronization issues in most cases


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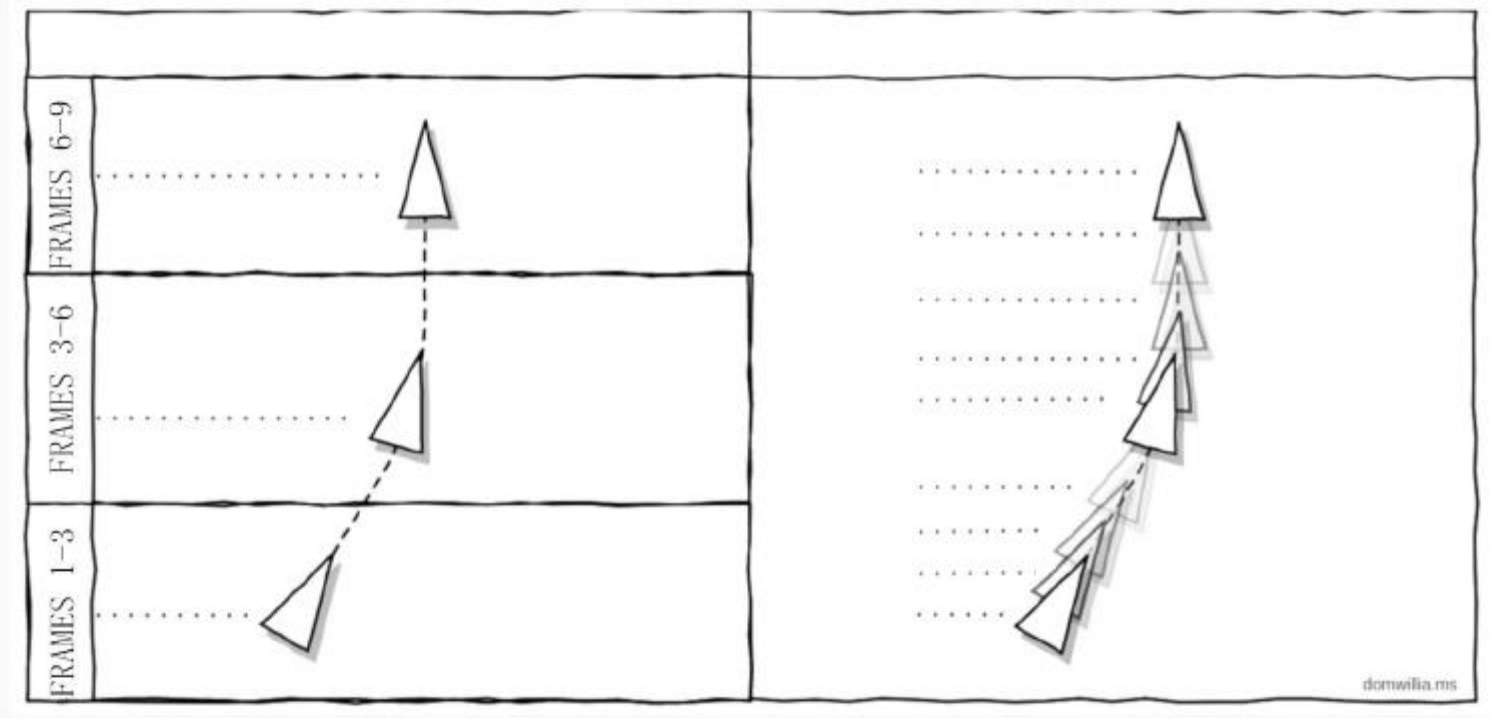
Latency Handling

- Problems:
  - Laggy input response: Client-side prediction
  - Abrupt movement of entities: Entity Interpolation
  - Non-expected reaction to inputs

Entity Interpolation

- Sometimes prediction is not possible (e.g. sudden changes in direction or speed)
  - Updating players positions on received data could lead to teleports (not good)

Solution: Interpolation position between 2 known points in time



Entity Interpolation

- Problems:
  - You are showing the player always in the past
  - Every player sees a different version of the movement
  - Could lead to seeing a player where it isn't
- Can be solved with more resolution, or with Lag Compensation (next)
- Implementation: have an initial (current) and a final (target) representation of the world state, every frame, interpolate among them

Latency Handling

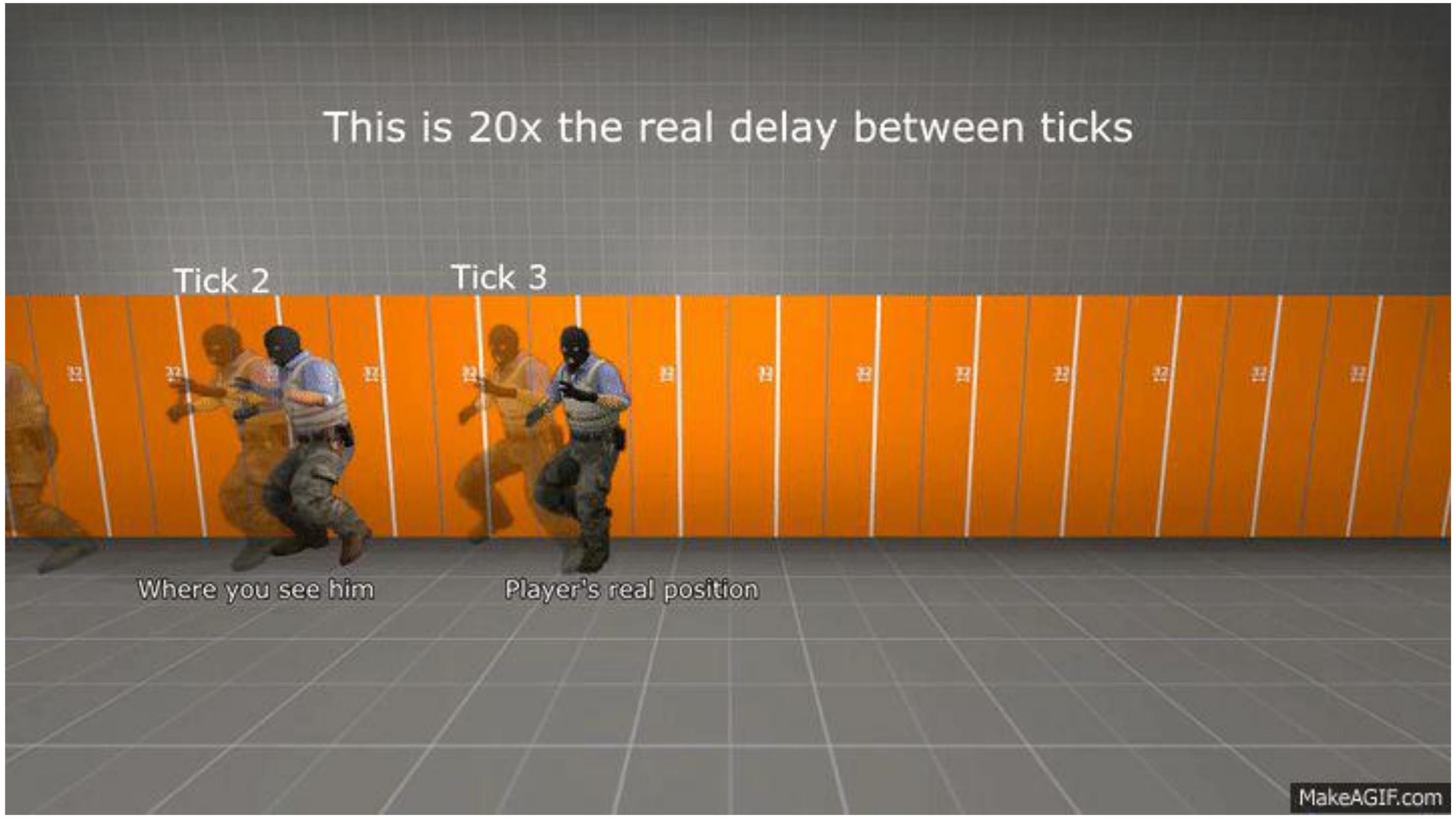
- Problems:
  - Laggy input response: Client-side prediction
  - Abrupt movement of entities
  - Non-expected reaction to inputs: Lag Compensation

Lag Compensation

- When you shoot, client sends this event to the server with full information: the exact timestamp of your shot, and the exact aim of the weapon.
- Here's the crucial step. Since the server gets all the input with timestamps, it can authoritatively reconstruct the world at any instant in the past. In particular, it can reconstruct the world exactly as it looked like to any client at any point in time.
- This means the server can know exactly what was on your weapon's sights the instant you shot. It was the past position of your enemy's head, but the server knows it was the position of his head in your present.

Lag Compensation

- The server processes the shot at that point in time, and updates the clients.



Additional Reading

- Entity interpolation https://www.gabrielgambetta.com/entity-interpolation.html
- Lag compensation https://www.gabrielgambetta.com/lag-compensation.html
- Client-side Prediction
- https://www.gabrielgambetta.com/client-side-prediction-server-reconciliation.html

Reminder

- In Atenea you have a file with functions to simulate Packet Loss and Jitter
- Your submission should include these functions and use them for sending packets across clients/servers.
- This will be used to evaluate the part in which you implement the compensation methods depicted in the last recent slides (Lab 8-9), in case you do